Tomer Gresler

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ABOUT

B.Sc. Computer Science. I like understanding a system from the bottom up, and building things people actually use. I wrote a **two-pass assembler in C** from scratch, a **client-server system** with its own binary protocol and **end-to-end encryption**, and a **REST API** in Python running in Docker. I don't stop when the code works; I **dig in until I find the real cause**. A **self-paced degree** taught me to **learn new technologies on my own**, and interviewing passengers in aviation security and serving on a Navy crew taught me to **communicate clearly and stay reliable** under pressure. I use **AI tools like Claude Code** as part of my daily workflow.

SKILLS

- **Languages:** C, Python
- **Systems & Networking:** Memory and pointers, TCP/IP sockets, binary protocol design, client-server architecture, Linux
- **Backend & Data:** REST APIs (FastAPI, Pydantic), SQL and relational database design (SQLite)
- **Security:** AES/RSA encryption, input validation, SQL injection prevention
- **Tools:** Git, GitHub Actions CI, Docker, GCC, Make
- **Engineering:** Data structures & algorithms, OOP, debugging, testing

PROJECTS

MessageU – Encrypted Messaging Client-Server *Python, C++, SQLite, Crypto++*

- Python server (selectors, SQLite) and C++ client talking over a custom binary TCP protocol: fixed headers, length-prefixed payloads, request and response codes.
- End-to-end encryption with RSA-OAEP key exchange and AES-CBC messages, so the server never sees the text.
- Found and fixed real networking bugs (partial reads, a reused closed socket, duplicated responses) and added socket-level tests with GitHub Actions CI.

Two-Pass Assembler *C, Linux, Make*

- Wrote, on my own and from scratch in C, an assembler that turns a custom assembly language into machine code.
- Macro expansion, a symbol table built as a linked list, and two passes that encode instructions into 14-bit words and resolve labels and external references.

WatchShelf – Movie Tracking REST API *Python, FastAPI, SQLite, Docker*

- REST API to search movies, track status, ratings and notes, and manage custom lists, with a relational SQLite schema (many-to-many join table).
- Pydantic input validation and parameterized SQL with a whitelisted ORDER BY to block SQL injection; containerized with Docker.

EDUCATION

B.Sc. Computer Science — The Open University of Israel 2023 – 2026

- GPA: 85 | Dean's Honor List (first year of study) — self-paced program with no mandatory lectures; I managed my own coursework, pace and deadlines

EXPERIENCE

Security Profiler, Israeli Airport Authority 2019 – 2021

- Interviewed passengers and made real-time risk decisions under pressure.
- Strict attention to detail and adherence to operational protocols, with no room for mistakes.

MILITARY SERVICE

Naval Combat Soldier (Missile Ship), Israel Defense Forces (Navy) 2016 – 2019

- Served as part of a ship's crew in demanding conditions, where everyone depends on everyone.

LANGUAGES

Hebrew (Native) | English (Fluent)